Apex Distributed Systems LLC

ENTERPRISE CLOUD ARCHITECTURE AND DISTRIBUTED INFRASTRUCTURE

100 Montgomery Street, Suite 2400
San Francisco, CA, 94104
Tax ID: US-84-9921048 | Email: invoicing@apex.systems

INVOICE

Invoice Number:	INV-2026-0047
Issue Date:	2026-10-11
Due Date:	2026-11-10
Payment Terms:	Net 30 Days
Currency:	USD

BILLED TO (CLIENT)

Vanguard Distributed Networks

Platform and Cloud Infrastructure Engineering
Enterprise Technology Plaza, Floor 12
Austin, TX, 10001
Attention: Marcus Vance, Chief Architect
Email: billing@vanguard.io

REMIT PAYMENT TO

JPMorgan Chase Bank, N.A.

Account Holder: Apex Distributed Systems LLC
Account Number: **** **** 7821
Routing Number (ACH): 021000021
SWIFT / BIC: CHASUS33
Reference: INV-2026-0047

#	SKU	DESCRIPTION	QTY	RATE (USD)	AMOUNT (USD)
1	ENG-INF-001	Distributed consensus state-machine replica verification and Byzantine audit (Milestone 1)	2	$480.00	**$960.00**
2	ENG-INF-002	High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 2)	3	$350.00	**$1050.00**
3	ENG-INF-003	Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 3)	4	$275.00	**$1100.00**
4	ENG-INF-004	Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 4)	1	$320.00	**$320.00**
5	ENG-INF-005	Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 5)	2	$450.00	**$900.00**
6	ENG-INF-006	Virtual thread concurrency soak probe under simulated 10k connections (Milestone 6)	3	$390.00	**$1170.00**
7	ENG-INF-007	Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 7)	4	$260.00	**$1040.00**
8	ENG-INF-008	Distributed consensus state-machine replica verification and Byzantine audit (Milestone 8)	1	$480.00	**$480.00**
9	ENG-INF-009	High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 9)	2	$350.00	**$700.00**
10	ENG-INF-010	Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 10)	3	$275.00	**$825.00**
11	ENG-INF-011	Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 11)	4	$320.00	**$1280.00**
12	ENG-INF-012	Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 12)	1	$450.00	**$450.00**

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#	SKU	DESCRIPTION	QTY	RATE (USD)	AMOUNT (USD)
13	ENG-INF-013	Virtual thread concurrency soak probe under simulated 10k connections (Milestone 13)	2	$390.00	**$780.00**
14	ENG-INF-014	Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 14)	3	$260.00	**$780.00**
15	ENG-INF-015	Distributed consensus state-machine replica verification and Byzantine audit (Milestone 15)	4	$480.00	**$1920.00**
16	ENG-INF-016	High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 16)	1	$350.00	**$350.00**
17	ENG-INF-017	Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 17)	2	$275.00	**$550.00**
18	ENG-INF-018	Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 18)	3	$320.00	**$960.00**
19	ENG-INF-019	Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 19)	4	$450.00	**$1800.00**
20	ENG-INF-020	Virtual thread concurrency soak probe under simulated 10k connections (Milestone 20)	1	$390.00	**$390.00**
21	ENG-INF-021	Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 21)	2	$260.00	**$520.00**
22	ENG-INF-022	Distributed consensus state-machine replica verification and Byzantine audit (Milestone 22)	3	$480.00	**$1440.00**
23	ENG-INF-023	High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 23)	4	$350.00	**$1400.00**
24	ENG-INF-024	Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 24)	1	$275.00	**$275.00**
25	ENG-INF-025	Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 25)	2	$320.00	**$640.00**
26	ENG-INF-026	Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 26)	3	$450.00	**$1350.00**
27	ENG-INF-027	Virtual thread concurrency soak probe under simulated 10k connections (Milestone 27)	4	$390.00	**$1560.00**
28	ENG-INF-028	Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 28)	1	$260.00	**$260.00**
29	ENG-INF-029	Distributed consensus state-machine replica verification and Byzantine audit (Milestone 29)	2	$480.00	**$960.00**
30	ENG-INF-030	High-throughput reactive event ingestion pipeline tuning and partition testing (Milestone 30)	3	$350.00	**$1050.00**
31	ENG-INF-031	Native PostgreSQL read-replica connection pooling optimization via PgBouncer (Milestone 31)	4	$275.00	**$1100.00**

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#	SKU	DESCRIPTION	QTY	RATE (USD)	AMOUNT (USD)
32	ENG-INF-032	Redis cluster cache stampede mitigation via probabilistic early expiration (Milestone 32)	1	$320.00	**$320.00**
33	ENG-INF-033	Zero-downtime schema migration verification for high-concurrency order ledger (Milestone 33)	2	$450.00	**$900.00**
34	ENG-INF-034	Virtual thread concurrency soak probe under simulated 10k connections (Milestone 34)	3	$390.00	**$1170.00**
35	ENG-INF-035	Distributed trace propagation audit and OpenTelemetry exporter alignment (Milestone 35)	4	$260.00	**$1040.00**

Subtotal:	**$31790.00**
Sales Tax (8.5%):	**$2702.15**
Shipping and Handling:	**$125.00**
Total Due:	**$34617.15**

TERMS AND CONDITIONS

Payment is due within Net 30 Days of invoice issuance. Late balances are subject to a 1.5% monthly finance charge. All distributed computing consulting services, infrastructure deliverables, and architectural implementations are certified under Enterprise SLA terms. For inquiries regarding this statement, please contact accounting@apex.systems.

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